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GitAIOps

A 4-Layer Architecture for Predictable AI-Assisted Operations

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Who is telling you this?



Hoon Jo

CNCF & AAIF Ambassador

advocating open source across cloud native and agentic AI

Author

wrote the book behind today's demo, and four more (AI, k8s, Ansible, Python)

AI & Cloud Native Engineer

ran the production migration in this talk, with one AI agent



Your AI agent has amnesia

Forgetting yesterday's decisions, values, and lessons: gone by the next session

Guesswork so it fills the gaps: re-decides settled choices, invents commands and flags

Unpredictability so the same request gives a different result: production cannot run on that

Every session starts from zero. The infrastructure does not.



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Git "AI" Ops

GitAIOps the AI joins GitOps with what it never had: a memory that lasts



The test bed: a real production migration

15

Helm releases

Kafka

ZooKeeper to KRaft

Valkey

replacing Redis

Rebuilt

observability stack

Constraint: the old platform keeps sending messages 24/7. No downtime allowed.

One AI agent, one operator, and a deadline.



Layer 1 work-plans: plans humans read

36 files

23,854 lines

Why + How + trade-offs

every component decision written down

Migration plan and runbooks

infrastructure, platform, data, transition

WHAT BROKE Hand all of this to the AI as-is, and it drowns in context.



Layer 2 claude-context: distilled for the AI

6 files

1,254 lines (19:1)

A project state dashboard

summary, environment values, execution guide

Textbook vs exam notes

work-plans are the textbook, this is the summary

WHAT BROKE The AI understood the state, then improvised its own commands.



Layer 3 command-guardrails: control the hands

117 files

step-by-step commands

Execution order enforced

00-prerequisites through 40-transition, then rollback

Old platform protected

read-only rules: get, describe, logs only

WHAT BROKE Right command, creative flags. Versions and values still drifted.



Layer 4 locked values: remove interpretation

30 files

versions and values pinned

Real incident

unpinned Mimir chart auto-upgraded 5.8.0 to 6.0.5, CrashLoop

Everything explicit

version, values file, namespace, timeout: all fixed

RESULT Same input, same deployment. Nothing left to improvise.



Four layers, four freedoms removed

Layer 1 work-plans

what to build and why

removes missing context

Layer 2 claude-context

current state, distilled

removes stale memory

Layer 3 command-guardrails

what to run, in order

removes improvised steps

Layer 4 locked values

exact versions and values

removes interpretation

Predictability is what remains when the freedoms are gone.



What predictability bought us

2 weeks → 2 days

full DEV environment build

1 week → 1 day

PROD build, guardrails reused

Zero downtime

old platform never stopped

The speed did not come from the AI typing faster.

It came from never having to redo an unpredictable step.



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Live Demo: Ask the Repository



The production repo? I cannot show you

Private

the migration repo stays behind NDA.
Numbers only.

Rebuilt in public

a book and notiflex-platform: the same
pattern, end to end.



gitaiops/
notiflex-platform

LIVE DEMO let's ask the repository three questions

1. Why Alertmanager over Grafana Alerting?
2. Onboard a new engineer.
3. Ask anything: your turn.





gitaioops/
notiflex-platform

Why Alertmanager over Grafana Alerting?

the repository answers from 16 recorded decisions



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Plan a new engineer's first week

a first-week checklist, generated from the repository



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notiflex-platform

Ask anything

your turn: the repository will answer

What you just saw

The repo answered why 16 recorded decisions, alternatives included: no asking around

An onboarding guide, generated derived from the repo. Nobody ever wrote it by hand

Your question, answered live, from the same recorded memory: nothing staged for it

Your repository became the single source of truth. Nobody wrote a wiki.



Make it yours: start small

- 1. Start now: record as you go** rules and current state, in whatever file your agent reads
- 2. Plan first when the work is big** a migration moves settled decisions: plan it, pin the values
- 3. Keep the single source of truth alive** everyone updates it, and the AI keeps it readable

Works with any coding agent. The architecture lives in Git, not in the tool.

The migration ends. The memory keeps working.



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Recap



GitAIOps in three sentences

Git

Git is the memory

every plan, command, and value lives in the repository

AI

AI is the operator

guided by a 4-layer architecture: context, guardrails, pinned values

Ops

Ops is predictable

AI-assisted operations from a single source of truth

Thank you!



GitAI Ops on GitHub
github.com/gitaiops



Hands-on with the book
github.com/gitaiops/_Book_GitAIops



Hoon Jo

CNCF & AAIF Ambassador

UPCOMING Aug 13

MCP Dev Summit Seoul, same venue

AAIF Ambassador AMA sched.co/2QScw

Book signing 15:15 sched.co/2TTsf